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# Problem Set III

1. Show that a GARCH(1,1) model can be written as an infinite ARCH process (assuming the weak stationarity condition holds). *Comment*: This explains why GARCH models are typically more empirically relevant than ARCH models.

$GARCH(1,1)$: $\epsilon_t = \sqrt{h_t} v_t, and\ h_t = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 h_{t-1}$

And:

$h_{t-1} = \alpha_0 + \alpha_1 \epsilon_{t-2}^2 + \beta_1 h_{t-2}$

$\Rightarrow h_t = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1(\alpha_0 + \alpha_1 \epsilon_{t-2}^2 + \beta_1 h_{t-2}) = (\alpha_0 + \alpha_0 \beta_1) + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \alpha_1 \epsilon_{t-2}^2 + \beta_1^2 h_{t-2}$

Doing this process recursively, we will have:

$$h_t = \alpha_0 \sum_{i=0}^{\infty} \beta_1^i + \alpha_1 \sum_{i=0}^{\infty} \beta_1^i \epsilon_{t-i-1}^2 = \sum_{i=0}^{\infty} \beta_1^i (\alpha_0 + \alpha_1 \epsilon_{t-i-1}^2)$$

Give that: $\alpha_1 + \beta_1 < 1, and\ \alpha_0, \alpha_1, \beta_1 > 0$:

$$h_t = \frac{\alpha_0}{1-\beta_1} + \sum_{i=0}^{\infty} \alpha_1 \beta_1^i \epsilon_{t-i-1}^2 = \omega + \sum_{i=0}^{\infty} \gamma_i \epsilon_{t-i-1}^2, which\ \omega = \frac{\alpha_0}{1-\beta_1}\ and\ \gamma_i = \alpha_1 \beta_1^i$$

Which is $ARCH(\infty)$.

2. Consider the model

$$\begin{aligned} y_t &= \phi_0 + \phi_1 y_{t-1} + \varepsilon_t, \\ \varepsilon_t &= \sqrt{h_t} v_t, \quad v_t \sim IID\,(0,1)\,, \\ h_t &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2, \end{aligned}$$

where $|\phi_1| < 1$, $\alpha_0 > 0$, $\alpha_i \geq 0$, for $i = 1, 2$, and $\alpha_1 + \alpha_2 < 1$.

(a) Derive the conditional mean and variance of $y_t$ given $\Omega_{t-1} = \{y_{t-1}, y_{t-2}, \ldots\}$.

(b) Derive the unconditional mean and variance of $y_t$.

(a)

$\mathbb{E}_{t-1}[y_t] = \mathbb{E}_{t-1}[\phi_0 + \phi_1 y_{t-1} + \epsilon_t] = \phi_0 + \phi_1 y_{t-1} + \mathbb{E}_{t-1}[\epsilon_t]$

And

$$\mathbb{E}_{t-1}[\epsilon_t] = \mathbb{E}_{t-1}\left[v_t\sqrt{\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2}\right] = \sqrt{\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2}\,\mathbb{E}_{t-1}[v_t] = 0$$

$$\Rightarrow \mathbb{E}_{t-1}[y_t] = \phi_0 + \phi_1 y_{t-1}$$

$$Var_{t-1}[y_t] = \mathbb{E}_{t-1}[(y_t - \mathbb{E}_{t-1}[y_t])^2] = \mathbb{E}_{t-1}[\epsilon_t^2] = h_t\mathbb{E}_{t-1}[v_t^2] = h_t = \alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2$$

Final Answer:

$$\mathbb{E}_{t-1}[y_t] = \phi_0 + \phi_1 y_{t-1}$$

$$Var_{t-1}[y_t] = h_t$$

(b)

$$\mathbb{E}[y_t] = \mathbb{E}\big[\mathbb{E}_{t-1}[y_t]\big] = \mathbb{E}[\phi_0 + \phi_1 y_{t-1}] = \phi_0 + \phi_1\mathbb{E}[y_{t-1}] \Rightarrow \mathbb{E}[y_t] = \frac{\phi_0}{1-\phi_1}$$

$$Var[y_t] = Var(\mathbb{E}_{t-1}[y_t]) + \mathbb{E}[Var_{t-1}(y_t)] = Var(\phi_0 + \phi_1 y_{t-1}) + \mathbb{E}[h_t]$$

Since we have $Var(y_{t-1}) = Var(y_t)$,

$$\mathbb{E}[h_t] = \mathbb{E}\big[\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2_t}^2\big] \Rightarrow \mathbb{E}[h_t] = \frac{\alpha_0}{1-\alpha_1-\alpha_2}$$

$$\Rightarrow Var[y_t] = \frac{\frac{\alpha_0}{1-\alpha_1-\alpha_2}}{1-\phi_1^2}$$

Final answer:

$$\mathbb{E}[y_t] = \frac{\phi_0}{1-\phi_1}$$

$$Var[y_t] = \frac{\frac{\alpha_0}{1-\alpha_1-\alpha_2}}{1-\phi_1^2}$$

3. Consider the GARCH(1,2) process $\varepsilon_t = \sqrt{h_t}v_t$, where $v_t \sim IID(0,1)$,

$$h_t = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \alpha_2\varepsilon_{t-2}^2 + \beta_1 h_{t-1},$$

$\alpha_0 > 0$, $\alpha_i \geq 0$, for $i = 1, 2$, $\beta_1 \geq 0$, and $\alpha_1 + \alpha_2 + \beta_1 < 1$.

(a) Show that $E\left(\varepsilon_t^2|\varepsilon_{t-1}, \varepsilon_{t-2}, \ldots\right) = h_t$.

(b) Derive the unconditional variance of $\varepsilon_t$.

(c) Derive the $h$-step ahead forecasts of $h_{t+h}$ at the forecast origin $t$, for $h \geq 1$.

(a)

$$\mathbb{E}_{t-1}[\epsilon_t^2] = \mathbb{E}_{t-1}[h_t v_t^2] = \mathbb{E}_{t-1}[(\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2 + \beta_1 h_{t-1})v_t^2]$$
$$= (\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2 + \beta_1 h_{t-1})\mathbb{E}_{t-1}[v_t^2] = h_t$$

(b)

$$\mathbb{E}[\epsilon_t^2] = \mathbb{E}\Big[\mathbb{E}_{t-1}[\epsilon_t^2]\Big] = \mathbb{E}[h_t]$$

And

$$\mathbb{E}[h_t] = \mathbb{E}[\alpha_0 + \alpha_1\epsilon_{t-1}^2 + \alpha_2\epsilon_{t-2}^2 + \beta_1 h_{t-1}] \Rightarrow \mathbb{E}[h_t] = \frac{\alpha_0}{1 - \alpha_1 - \alpha_2 - \beta_1}$$

$$\Rightarrow \mathbb{E}[\epsilon_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \alpha_2 - \beta_1}$$

$$\mathbb{E}[\epsilon_t] = \mathbb{E}\big[\mathbb{E}_{t-1}[\epsilon_t]\big] \; and \; \mathbb{E}_{t-1}[\epsilon_t] = \sqrt{h_t}\;\; \mathbb{E}[v_t] = 0 \Rightarrow \mathbb{E}[\epsilon_t] = 0$$

$$Var(\epsilon_t) = \mathbb{E}[\epsilon_t^2] - \mathbb{E}^2[\epsilon_t] = \mathbb{E}[\epsilon_t^2] = \frac{\alpha_0}{1 - \alpha_1 - \alpha_2 - \beta_1}$$

(c)

$$\hat{h}_{t+1|t} = \mathbb{E}_t[h_{t+1}] = \alpha_0 + \alpha_1\epsilon_t^2 + \alpha_2\epsilon_{t-1}^2 + \beta_1 h_t$$

$$\hat{h}_{t+2|t} = \alpha_0 + \alpha_1\epsilon_{t+1}^2 + \alpha_2\epsilon_t^2 + \beta_1\hat{h}_{t+1|t} = \alpha_0 + \alpha_1\mathbb{E}[\epsilon_{t+1}^2|\Omega_t] + \alpha_2\epsilon_t^2 + \beta_1\hat{h}_{t+1|t}$$
$$= \alpha_0 + \alpha_2\epsilon_t^2 + (\alpha_1 + \beta_1)\hat{h}_{t+1|t}$$

$$\hat{h}_{t+3} = \alpha_0 + \alpha_1\epsilon_{t+2}^2 + \alpha_2\epsilon_{t+1}^2 + \beta_1\hat{h}_{t+2|t} = \alpha_0 + \alpha_1\hat{h}_{t+2|t} + \alpha_2\hat{h}_{t+1|t} + \beta_1\hat{h}_{t+2|t}$$
$$= \alpha_0 + \alpha_2\hat{h}_{t+1|t} + (\alpha_1 + \beta_1)\hat{h}_{t+2|t}$$

$$\hat{h}_{t+h} = \alpha_0 + \alpha_2\hat{h}_{t+h-2|t} + (\alpha_1 + \beta_1)\hat{h}_{t+h-1|t} \;\; for\; h > 2$$

This recurrence is a linear difference equation of order 2. The homogeneous part is:

$$x_h = (\alpha_1 + \beta_1)x_{h-1} + \alpha_2 x_{h-2}$$

Which has characteristic function as below:

$\lambda^2 - (\alpha_1 + \beta_1)\lambda - \alpha_2 = 0$ which gives us the homogeneous solution of

$c_1\lambda_1^h + c_2\lambda_2^h$ and since the $\alpha_1 + \beta_1 + \alpha_2 < 1$ and $\alpha_i > 0, \beta_i > 0$, then:

$\Rightarrow |\lambda_1|, |\lambda_2| < 1$

And for the particular solution I am peaking the steady state which is:

$$\frac{\alpha_0}{1-\alpha_1-\alpha_2-\beta_1}$$

So:

$$\hat{h}_{t+h} = \frac{\alpha_0}{1-\alpha_1-\alpha_2-\beta_1} + c_1\lambda_1^h + c_2\lambda_2^h$$

Therefore, as $h \to \infty$ then $\hat{h}_{t+h} \to \frac{\alpha_0}{1-\alpha_1-\alpha_2-\beta_1}$

$$\Rightarrow \hat{h}_{t+h} = \alpha_0 \frac{(1-(\alpha_1+\beta_1)^{h-2})}{1-(\alpha_1+\beta_1)} + \alpha_2 \frac{(1-(\alpha_1+\beta_1)^{h-2})}{1-(\alpha_1+\beta_1)} + (\alpha_1+\beta_1)^{h-2}\hat{h}_{t+2|t}$$

4. Consider the model for $\mathbf{y}_t = (y_{1t}, y_{2t})'$:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t})' \sim WN(\mathbf{0}, \boldsymbol{\Sigma})$ with

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1.0 & 0.7 \\ 0.7 & 1.5 \end{pmatrix}.$$

(a) Is the bivariate series $\mathbf{y}_t$ weakly stationary?

(b) Find the mean vector of $\mathbf{y}_t$.
*Hint*: The inverse of a $2 \times 2$ invertible matrix $\mathbf{A}$ has the formula:

$$\text{If } \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \text{ then } \mathbf{A}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

(c) Suppose the sample size is $T = 100$, and in the last period we observed $y_{1,100} = 1$ and $y_{2,100} = 0.5$. What are the forecasts of $y_{1,101}$ and $y_{2,101}$ and their associated standard errors?

(d) Find the coefficient matrices on $\boldsymbol{\varepsilon}_t$, $\boldsymbol{\varepsilon}_{t-1}$ and $\boldsymbol{\varepsilon}_{t-2}$ in the MA representation of $\mathbf{y}_t$.

(a)

For check the stationarity of our VAR model we should check the eigenvalue of our matrix, and they need be less than one in modulus:

$$\Phi = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}$$

The eigen value comes from:

$$|\Phi - \lambda I| = 0 \Rightarrow \begin{vmatrix} 0.2 - \lambda & 0.3 \\ -0.6 & 1.1 - \lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 = 0.8, \lambda_2 = 0.1$$

Therefore, the bivariate $y_t$ is weakly stationary.

(b)

$$\mathbb{E}[y_t] = (I - \Phi)^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 - 0.2 & -0.3 \\ 0.6 & 1 - 1.1 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.8 & -0.3 \\ 0.6 & -0.1 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -8 \end{pmatrix}$$

(c)

Assuming that: $\mathbb{E}[\epsilon_t] = 0$

$$\hat{y}_{101} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix} \begin{pmatrix} 1 \\ 0.5 \end{pmatrix} = \begin{pmatrix} 4.35 \\ 1.95 \end{pmatrix}$$

$$Var(y_t) = (1 - \Phi)^{-1}\Sigma = \begin{pmatrix} 1 - 0.2 & -0.3 \\ 0.6 & 1 - 1.1 \end{pmatrix}^{-1} \begin{pmatrix} 01 & 0.7 \\ 0.7 & 1.5 \end{pmatrix} = \begin{pmatrix} 1.1 & 3.8 \\ -0.4 & 7.8 \end{pmatrix}$$

So, the standard errors are: $\sqrt{1.1} = 1.04889 \; and \; \sqrt{7.8} = 2.792848$

(d)

$$\bar{y}_t = \epsilon_t + \Phi\epsilon_{t-1} + \Phi^2\epsilon_{t-2} + \cdots$$

So,

Coefficient of $\epsilon_t$ is $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Coefficient of $\epsilon_{t-1} = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}$

Coefficient of $\epsilon_{t-2} = \begin{pmatrix} 0.2 & 0.3 \\ -0.6 & 1.1 \end{pmatrix}^2 = \begin{pmatrix} -0.14 & 0.39 \\ -0.78 & 1.03 \end{pmatrix}$